

Replication Package

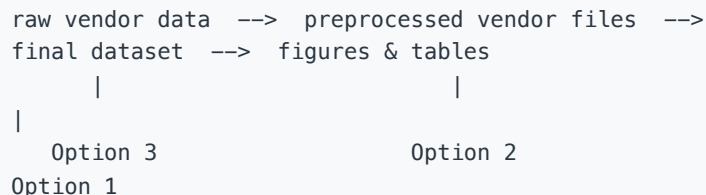
Constrained Investors in the Corporate Bond Markets - Anticipation of Credit Rating Downgrades Matthias Xaver

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This package reproduces all figures and tables in the paper. It builds a disaggregated bond $\times$ fund $\times$ firm $\times$ quarter panel from eMAXX, Mergent FISD, WRDS Bond Returns, S&P Capital IQ, and Markit CDS, then runs descriptive statistics and stacked difference-in-differences event studies around the first credit rating downgrade. It is organized so a reproduction team can choose **how far upstream to start**.

How to reproduce

Goal: regenerate every figure and table in the paper. The data moves through three stages, and you choose which stage to start from:



Edit the **two settings at the top of** `setup.do` — `${REPL}` (where this package lives on your machine) and `${mode}` (which stage you start from) — then **run** `setup.do`. The three options are fully independent: each writes to its own location and leaves the others untouched, so you can run any one of them — or all three — without them affecting each other.

Option 1 — from the final, fully processed dataset · ≈ 20 minutes

`mode = "shipped"` (the default). Start from the finished dataset that ships with the package and simply produce the figures and tables. Fastest path — there is nothing to build.

Option 2 — from the preprocessed vendor data · ≈ 1–2 hours

`mode = "reference"` . Start from the cleaned, per-vendor files: merge them and run the sample selection yourself to rebuild the final dataset, then produce the figures and tables.

Option 3 — from the raw vendor data, from scratch · several hours

`mode = "raw"` . Start from the original raw vendor downloads: rebuild each preprocessed vendor file, then do the merge and sample selection, then the figures and tables — the complete end-to-end reproduction.

What to run

1. `setup.do` — set `${REPL}` and `${mode}` , then run it.
2. `Sample Replication/0_run_sample.do` — rebuilds the dataset. Needed for Options 2 and 3 only; for Option 1 there is nothing to build, so skip it.
3. `Paper Replication/0_run_paper.do` — writes every figure and table to `Paper Replication/Figures and Tables/` .

The preprocessed vendor data — do not overwrite

These are the cleaned, per-vendor files that **Option 2** reads. They reproduce the **exact** paper findings:

```
Data/Reference Files/CapitalIQ_Final.dta
Data/Reference Files/CDS_2012_2020_GVKEY-CUSIP.dta
Data/Reference Files/WRDS_Bond>Returns.dta
Data/MergentFISD/Paper Reference
File/FINALIssueRatings.dta ← rich MergentFISD panel
(+ its build .do)
```

Option 2 (`reference`) reads these directly. **Option 3** (`raw`) regenerates its own copies from the raw vendor data in `Data/<source>/` instead (for MergentFISD, `MergentFISD_QuarterlyPanel_2012–2023.dta`). **Option 1** (`shipped`) reads neither — it uses the finished dataset directly. Sample construction only ever writes to `Data/<source>/` and `Data/Working Files/` — it never touches the files above.

Kept read-only: these files ship read-only and `setup.do` re-asserts the lock on every run, so a rebuild — even a from-scratch one — cannot overwrite them; the OS denies the write. (To deliberately refresh one, `chmod u+w` it first.)

Structure

```
setup.do                ← edit ${REPL} once;
defines all paths + the reference/raw choice
README.md
VARIABLE_DICTIONARY.md  ← definitions of key
constructed variables (audit aid)

Sample Replication/      ← DATA CONSTRUCTION
  0_run_sample.do        ← orchestrator: (raw
only) source builds → Sample_Creation
  Sample_Creation.do     ← builds Working
Files/SampleFinalCDS(.dta), _WV(.dta)
  Build_Master.do        ← builds Working
Files/_master.dta (run by the paper orchestrator)
  Merge_Variables.do     ← reference catalog of
optional merges

Paper Replication/       ← ANALYSIS
  0_run_paper.do         ← orchestrator:
Build_Master → all analysis do-files
  Code/                  ← 1_Descriptives,
2_Baseline_Analysis, 2c_, 3_, UPGRADE, Robustness
  Figures and Tables/    ← output (one subfolder
per section + Tables and Figures in Paper/)

Investment Management Sample Creation/ ← OPTIONAL:
joinby eMAXX personnel onto the sample

Data/                    ← all data; each source
folder holds its OWN build do-file + Raw Data/
  eMAXX/                  (eMAXX_1998-2023.do + Raw eMAXX/ +
*_Complete.dta)
  MergentFISD/            (MergentFISD.do + Raw Data/ + rating
panels + IssuesLookup;
                        Paper Reference File/ =
FINALIssueRatings.dta + its build .do)
  CapitalIQ/              (CapitalIQ_DO.do + Raw Data/ +
CapitalIQ_Final)
  WRDS Bond Returns/      (WRDS_Bond_Returns.do + Raw Data/)
  Markit/                 (CDS Data.do + Raw Data/ + CDS panel)
  Working Files/          (shipped SampleFinalCDS, _WV,
_master; rebuilds → Rebuilt_reference/ | Rebuilt_raw/)
  Reference Files/        (preprocessed, read-only:
CapitalIQ_Final, CDS, WRDS_Bond_Returns)
```

Data on GitHub: the `Data/` tree is ~100 GB and is *not* committed (it is git-ignored). The repository holds the code, `setup.do`, and docs; the data is distributed separately (Dropbox / institutional access).

Pipeline map

What each script reads, writes, and in which mode it runs.

`${wsdir}` is the working-sample folder chosen by `${mode}` (Data/Working Files/ for shipped, `.../Rebuilt_reference/` or `.../Rebuilt_raw/` for a rebuild).

Step	Script	Reads	Writes	Runs in
0	setup.do	—	sets <code>\${REPL}</code> , <code>\${mode}</code> , <code>\${wsdir}</code> , all paths	always (first)
1	Sample Replication/0_run_sample.do	(raw) Data/<source>/Raw ...	orchestrates 1a + 1b	reference, raw
1a	eMAXX_1998–2023.do , MergentFISD.do , CapitalIQ_D0.do , WRDS_Bond>Returns.do , CDS Data.do	raw vendor files	Data/<source>/*_Complete / panels	raw only
1b	Sample Replication/Sample_Creation.do	<code>\${in_*}</code> vendor inputs + eMAXX *_Complete	<code>\${wsdir}/SampleFinalCDS.dta , ..._WV.dta</code>	reference, raw
2	Paper Replication/0_run_paper.do	<code>\${wsdir}/..._WV.dta</code>	orchestrates 2a–2e	always
2a	Sample Replication/Build_Master.do	<code>\${wsdir}/..._WV.dta</code>	<code>\${wsdir}/_master.dta</code>	always (via paper)
2b	Code/1_Descriptives.do	<code>\${wsdir}/_master.dta</code>	Descriptive/ → F01, F02, F03, T01, T02	always
2c	Code/2_Baseline_Analysis.do	<code>_master.dta</code>	Baseline_Analysis/ → T03–T07	always
2d	Code/2c_Baseline_Analysis_Extensive.do	<code>_master.dta</code>	Baseline_Analysis_Extensive/ → F04	always
2e	Code/3_Disaggregated_Threshold.do	<code>_master.dta</code>	Threshold/ → F05, F06	always
3	Investment Management Sample Creation/0_run_im_sample.do	<code>\${wsdir}/..._WV.dta</code> + <code>\${emaxx}/PERSONNEL_Complete.dta</code>	<code>\${wsdir}/..._IM_Sample.dta</code>	optional, any mode

In `shipped` mode step 1 builds nothing (the prebuilt sample is used); start at step 2. See `VARIABLE_DICTIONARY.md` for the variables these scripts construct, and `Paper Replication/README.md` for the internal-vs-paper table/figure numbering.

Requirements

Stata 17+ (tested on StataNow/MP). `setup.do` auto-installs **the required user-written commands** on its first run (only the ones that are missing), so no manual setup is needed — as long as the machine has internet access. For reference (or to pre-install offline), they are:

```
ssc install reghdfe
ssc install ftools
ssc install estout
ssc install coefplot
ssc install winsor2
ssc install egenmore
ssc install distinct
ssc install unique
net install grc1leg,
from("http://www.stata.com/users/vwiggins")
```